

No escape from syntax: Gikūyū nominalizations and the Complex Head analysis
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Mugane (1997) identifies two types of individual-denoting nominalizations in Gĩkũyũ (Bantu): the [mu...a]-type and the [mu...i]-type. He argues that the [mu...i]-type nominalizations are phrasal and that the [mu...a]-type nominalizations exhibit a puzzling nature, displaying both lexical and syntactic properties. This study examines Mugane's characterization, revisiting the notion of a lexicon-syntax divide. Applying Wood's (2023) Complex Head analysis, I demonstrate that we can explain the [mu...a]-type nominalizations within a syntactic framework without resorting to the lexicon. The analysis reveals that the puzzle is resolvable, and that syntax can account for both types of nominalizations in Gĩkũyũ.

Keywords: Nominalizations; Complex Head analysis; Phrasal Layering analysis; Gĩkũyũ

1 Introduction

Grammatical categories such as verbs (e.g., *to sing_V*) and nouns (e.g., *a song_N*) are universally attested. Quite interestingly, nouns can be derived from verbs (e.g., *a [[sing]_V-er]_N*). This process is referred to as nominalizations. Nominalizations have drawn interest since Lees (1960) due to their mixed display of verbal and nominal features (see also Chomsky 1970). Building on prior syntactic work (Abney 1987, Marantz 1997, Harley & Noyer 1998, Van Hout & Roeper 1998, Alexiadou 2001), I posit that deverbal nominalizations are derived from a nominal projection that includes some degree of verbal structure (Grimshaw 1990, Borsley & Kornfilt 1999, Kornfilt & Whitman 2011, Wood 2023), as sketched in (1):

- (1) [... *n* ... [... *v* ...]]

Although debate continues over how much verbal structure such nominalizations can exhibit, it is widely accepted that this varies across languages and constructions. Notably, most research has focused on event nominalizations.

In recent years, the structural variation in individual-denoting nominalizations has come to light (Baker & Vinokurova 2009, Fábregas 2012, Ntelitheos 2012, Toosarvandani 2014, Roy & Soare 2014, 2020, Harley 2020, Hanink 2021, Gotah & Lee 2024, Lee & Ndapo 2025).

This work focuses on Gĩkũyũ (Bantu) nominalizations in which a verb is realized together with a noun class prefix and a suffix spelled out as either *-i* or *-a*:

- (2) (a) **mu-end-i** a-ndu
 1-lover-I 2-person
 ‘a lover of people’
- (b) **mu-end-a** a-ndu
 1-lover-A 2-person
 ‘a lover of people’ (Mugane 1997:128, 131)

Mugane (1997) presents empirical facts from Gĩkũyũ nominalizations that cannot be readily handled under a traditional definition of syntax which assumes phrase-level constituents (see Alexiadou & Schäfer’s (2010) Phrasal Layering analysis):

- (3) “... [N]ouns marked with [mu-...-a] morphology present something of a paradox in that they exhibit lexical properties while still maintaining some phrasal characteristics ... [P]hrases cannot be syntactically attached to words. Yet such an attachment is precisely what we find with the formation of [mu-...-a] expressions, apparently calling to question the lexicon/syntax divide.” (Mugane 1997:127, 129)

Gĩkũyũ [mu-...-a]-type nominalizations are characterized as “phrases that are in the process of becoming lexicalized,” and “a lexicalization process involving phrases” (Mugane 1997:127). Despite such characterizations, I present a syntactic approach to handling this type of nominalizations without resorting to the lexicon. Adopting Wood’s (2023) Complex Head analysis, which runs on the assumption that a complex head can be derived via External Merge (EM), *without* head movement, I show that Gĩkũyũ [mu-...-a]-type nominalizations need not rely on

the inner workings of the lexicon.

This paper is organized as follows: Section 2 presents the empirical puzzle surrounding the [mu-...-a]-type nominalizations. Section 3 provides how the Complex Head analysis (Wood 2023) works and how it addresses the issue. Section 4 concludes.

2 Big and small nominalizations

2.1 Basic facts

Gĩkũyũ exhibits two types of nominalizations that differ with respect to their syntactic profile. The [mu-...-i]-type nominalizations permit various syntactic properties. The [mu-...-a]-type nominalizations, on the other hand, are quite limited with respect to what they can accommodate. The former is syntactically bigger than the latter. Hence, I refer to [mu-...-i]-type nominalizations as *big* nominalizations (BNs) and [mu-...-a]-type nominalizations as *small* nominalizations (SNs). (4) shows that an adverb can be realized inside BNs but not in SNs.

- (4) (a) a-thĩnj-**í** mbũri **ũũru**
 2-slaughter-I goats **badly**
 ‘those who slaughter goats badly’ (BN, Baker & Vinokurova 2009:547)
- (b) *mũ-thĩnj-**a** mbũri **ũũru**
 1-slaughter-A goats **badly**
 Intended: ‘one who slaughters goats badly’ (SN, Mugane 1997:133)

Additionally, an applied argument, together with its applicative marker *-er*, can be introduced in BNs but not in SNs:

- (5) (a) mũ-end-**er-i** andũ **mbiya**
 1-like-APPL-I 2.people **10.money**
 ‘one who likes people for money’ (BN, Mugane 1997:133)
- (b) *mũ-end-**er-a** andũ **mbiya**
 1-like-APPL-A 2.people **10.money**
 Intended: ‘one who likes people for money’ (SN, Mugane 1997:132)

Note that BNs can host a ditransitive predicate together with its internal arguments (IAs), unlike SNs:

- (6) (a) **mũ-tũm-i marũa thukuru**
 1-send-I 5.letter 9.school
 ‘a sender of a letter to a school’ (BN)
- (b) ***mũ-tũm-a marũa thukuru**
 1-send-A 5.letter 9.school
 Intended: ‘a sender of a letter to a school’ (SN, Mugane 1997:131)

BNs can also host the reflexive marker *ĩ-*, whereas SNs cannot:

- (7) (a) **mũ-ĩ-end-i**
 1-REFL-like-I
 ‘one who likes herself/himself’ (BN)
- (b) ***mũ-ĩ-end-a**
 1-REFL-like-A
 Intended: ‘one who likes herself/himself’ (SN, Mugane 1997:132)

Baker & Vinokurova (2009) argue that the reflexive marker *ĩ-* in Gĩkũyũ has to be locally bound by an antecedent. They assume that the external argument (EA), PRO, is introduced in the derivation and that it is the antecedent that locally binds the object reflexive marker. It is difficult to make such a case for SNs where the reflexive marker cannot be introduced.

Due to the various discrepancies observed between BNs and SNs, Mugane (1997) comes to the conclusion that SNs resemble lexical words.

2.2 The puzzle

Although BNs and SNs differ in various ways, they align in their ability to accommodate a phrasal theme argument (a theme DP). (8) demonstrates this point, which, in fact, complicates the picture sketched out in Section 2.1.

- (8) (a) **mũ-end-i [andũ aya othe anene a gĩchagi]**
 1-love-I 2.people 2.DEM 2.all 2.big 2.ASSOC 7.village
 ‘a lover of all these big village people’ (BN)
- (b) **mũ-end-a [andũ aya othe anene a gĩchagi]**
 1-love-A 2.people 2.DEM 2.all 2.big 2.ASSOC 7.village
 ‘a lover of all these big village people’ (SN, Mugane 1997:149)

The well-formedness of (8b) puts Mugane in a position to argue that SNs are derived only when syntax comes into play. There is a dilemma here since SNs

	Big nominalizations (BNs)	Small nominalizations (SNs)
Adverb	✓	✗
Applicative	✓	✗
Ditransitive	✓	✗
Reflexive marker	✓	✗
Theme DP	✓	✓

Table 1: (Im)possible syntactic properties in big and small nominalizations

do not display any other syntactic properties that are otherwise expected from a run-of-the-mill verbal or clausal structure. Table 1 summarizes our findings so far.

Mugane’s puzzle can be summarized as follows: how can something non-syntactic (e.g., an SN) accommodate something syntactic (e.g., a DP)? Further, how is the theme DP syntactically licensed in (8b) if syntactic operations do not apply SN-internally? In what follows, I address these questions.

3 Towards a solution

3.1 Phrasal Layering analysis

The Phrasal Layering analysis puts forward a unified approach to handling the clausal domain and the nominal domain. It works under the assumption that phrases such as ν P, VoiceP, and Asp(ect)P can be realized inside noun phrases (Alexiadou & Schäfer 2010, Ntelitheos 2012, Alexiadou 2020, Harley 2020, Hanink 2021).¹ While a unified approach to handling the two domains is theoretically desirable, the Phrasal Layering analysis is challenged by some important differences between the domains. The mismatch in Case distribution, for instance, sets the stage for the conundrum at play. It has been a long-standing puzzle why *of*-insertion is only observed in the nominal domain and not in the clausal domain (e.g., a *reader* **(of)* *many books* vs. *Mary reads* **(of)* *many books*)

[1] Relatedly, Gotah & Lee (2024) argue that NegP can be realized in Ewe (Kwa) individual-denoting nominalizations, and Lee & Ndapo (2025) claim that TP can be realized in Oshiwambo (Bantu) individual-denoting nominalizations. Quite interestingly, Tyler (2023) argues that English *-er* nominals can accommodate CP.

(see Chomsky 1970, Harley & Noyer 1998, Lee 2024). Moreover, in most Indo-European languages, adverbs are readily available in the clausal domain (e.g., *Halima reads books quickly.*), while they are rarely observed inside individual-denoting nominalizations (e.g., **a reader of books quickly.*). Analyses drawing parallels between the two domains have not been free from these issues.

3.2 Complex Head analysis

Taking into consideration the differences between the clausal domain and the nominal domain, Wood (2023) argues for a structure that does not involve phrase-level constituents inside nominalizations. This accounts for the presence of *of*-insertion and the absence of adverbs. Wood's structure is similar to a complex head established via head movement. At the heart of Wood's proposal is that complex heads need not rely on head movement. (9) schematizes Wood's way of deriving a complex head. The structure is deprived of specifiers and adjuncts.² We will soon see that this small syntax approach applies to SNs but not BNs.

(9) Deriving a complex head via EM (*without* head movement)

- (a) A root enters the derivation

$$\sqrt{\text{ROOT}}$$

- (b) $\sqrt{\text{ROOT}}$ and v undergo EM

$$\begin{array}{c} v \\ \diagup \quad \diagdown \\ \sqrt{\text{ROOT}} \quad v \end{array}$$

- (c) v and n undergo EM

$$\begin{array}{c} nP \\ \diagup \quad \diagdown \\ v \quad n \\ \diagup \quad \diagdown \\ \sqrt{\text{ROOT}} \quad v \end{array}$$

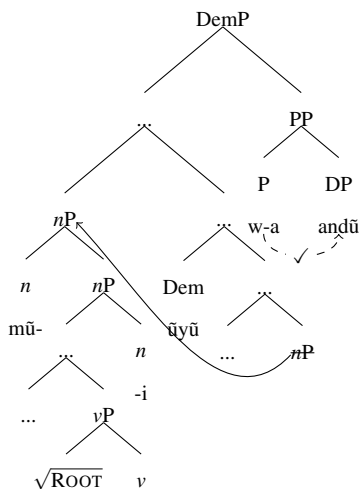
[2] This type of approach has been applied to nominalizations in Icelandic (Wood 2023) and German (Benz 2023) as well as to stative passives in English (Embick 2023, Biggs & Embick 2025), Greek (Paparounas 2023), and Ardalani Kurdish (Alex Hamo & Saman Meihami p.c.).

Crucially, (11) suggests that a theme DP can be introduced at different syntactic sites. In (11a), the DP is introduced *outside* of the verbal structure. In (11b), the DP is introduced *inside* of the verbal structure. Hence, the height at which the DP is introduced is crucial under the current analysis.

3.4 An analysis based on nominal licensing

I adopt the widely held assumption that all overt DPs have to be licensed in syntax.³ I take the associative marker in (11a) as a preposition (P) that licenses its complement DP in the nominal domain, as schematized in (12). Following Kramer (2015), Lee & Lee (2019), Fuchs & van der Wal (2022), I argue that a stacked-*n* analysis applies to the Gīkūyū nominalizations under discussion. Specifically, I posit that a noun class prefix belongs to *n*, which is stacked above a lower *n*, namely the suffix *-i*. In (12), movement is assumed to capture the correct word order between the noun and the demonstrative.

(12) Nominal licensing in (11a) (=BN)



[3] Sigurðsson (2012) argues that nominal licensing can be broken down into various relations such as thematic licensing, ϕ -licensing, and Case assignment. I take Nie's (2020, 2024) view that nominal licensing is necessary but need not involve Case assignment. Nie reports that many languages tolerate Case-less nominals, which aligns with Diercks' (2012) claim about Bantu languages. In light of this discussion, I am open to the idea that Gīkūyū need not involve Case assignment so long as nominal licensing is accomplished in some shape or form.

A reviewer asks how agreement is achieved between the associative marker and the noun class prefix in (12). I adopt Clem's (2023) probe-goal system. Her approach is based on Rezac's (2003) cyclic expansion analysis, which posits that probing can be done by an intermediate-level projection when the head fails to find an appropriate goal within its c-command domain. Clem argues that a maximal projection of X can also probe for a goal. Probing under this definition operates cyclically, and the search domain continues to expand whenever X and its kin(s) fail to find their goal. The expansion stops when X projects maximally (XP). Let us see how this works in (12). P is the probe and the upper *n* is the goal. P fails to probe the upper *n* since it is outside of P's c-command domain. On the second try, P's maximal projection probes its c-command domain, and this time the domain includes the upper *n* associated with the noun class. Agreement is successful on this trial, and the associative marker on P ends up with the noun class agreement marker.

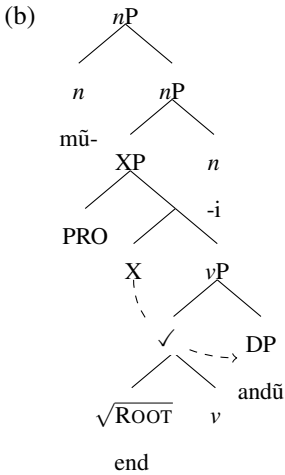
In cases where the associative marker is absent, the theme DP is introduced as the complement of the verb, and the DP is licensed inside the verbal structure. For present purposes, I assume that there is a higher functional head, call it X, that licenses the DP. Recall that the reflexive marker *ĩ-* can be showcased inside BNs, as seen in (7a). This suggests that an antecedent is present in the derivation. Adopting Baker & Vinokurova (2009), I assume that PRO takes part in BNs' derivation as an EA and acts as the antecedent for the reflexive marker *ĩ-*. For ease of exposition, I posit that the licensing of the theme DP is done in the style of Burzio's Generalization (Burzio 1986)⁴, though I remain open to alternative analyses. Note that the generalization is useful for cases where the direct object (DO) is present in the absence of a preposition. This is because the licensing of the DO can be done independently, as long as an EA is present in the derivation, as shown in (13).

[4] Below is Burzio's Generalization (Burzio 1986:178):

- (1) All and only the verbs that can assign a theta-role to the subject can assign accusative Case to an object.

- (13) (a) mũ-end-i a-ndũ
 1-love-I 2-person
 ‘a people lover’

(BN, Mugane 1997:128)



I argue that SNs bear an associative marker. I posit that *-a* is the associative marker (P) that licenses a theme DP.⁵ The PP led by *-a* here is sandwiched between the upper *nP* and the lower *nP*, as shown in (14). Since the PP participates in the derivation prior to the introduction of the upper *n* (noun class 1), the noun class agreement marker, *w-*, which would have otherwise appeared, does not show up. This suggests that an interpretable noun class feature-bearing goal (*n*) does not undergo agreement with its probe when the former c-commands the latter. In fact, this aligns with the idea that upward agreement is not desirable (see Chomsky 2000, 2001, Carstens 2005, Preminger 2014). I also take the view that agreement can fail without causing a crash in the derivation (Preminger 2014). As opposed to the verbal profile of a BN, the verbal profile of an SN lacks the necessary functional material to license a theme DP. This is well in line with the fact that a theme DP requires *-a* inside SNs. Here, I posit that there are two types of lower *ns*. One of them selects a phrase-level constituent (XP) and the other a head-level

[5] The idea that SNs accommodate an associative marker is in harmony with the view that *of*-insertion is observed in ‘small’ nominalizations (see Wood 2023).

fashion different from an ordinary XP containing verbal structure. Consequently, the phenomena at issue can be addressed in syntactic terms.

4 Conclusion

Mugane (1997) presents two types of nominalizations in Gikũyũ: BNs and SNs. Mugane argues the SNs resort to the lexicon and its inner workings prior to syntactic operations. I have highlighted that we can handle this issue without resorting to the lexicon. By making use of Wood's (2023) small syntax approach, I have argued that the facts about SNs can be captured simply by employing syntactic apparatuses. In doing so, the derivation is done within a single workspace. The approach advocated in this work also has cross-linguistic and cross-phenomenal implications. It adds weight to the recent growing body of literature on the Complex Head analysis, which has been applied for Icelandic and German nominalizations (Wood 2023, Benz 2023, 2025) and English and Greek stative passives (Embick 2023, Biggs & Embick 2025, Paparounas 2023). Hopefully, future research will shed further light on the exact profile of small syntax and its theoretical implications.

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